

Stabilizing lithium manganese oxide/organic carbonate electrolyte interface with a simple boron-containing additive



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ABSTRACT

A simple boron-containing molecule, tripropyl borate (TPB), is used as an electrolyte additive to stabilize the interface between spinel lithium manganese oxide (LiMn_2O_4) and carbonate-based electrolyte under elevated temperature. Electrochemical measurements indicate that the cyclic stability of LiMn_2O_4 electrode can be significantly improved by TPB. The capacity retention of LiMn_2O_4 at 1C after 200 cycles under 55 °C is improved from 47% to 74% by adding 3% TPB into a standard electrolyte (1.0 mol L⁻¹ LiPF₆-EC/EMC/DEC (3/5/2, in weight)). Most importantly, the self-discharge of LiMn_2O_4 under 55 °C, which takes place dramatically in the standard electrolyte, is effectively suppressed in 3% TPB-containing electrolyte. Theoretical calculations and physical characterizations demonstrate that a protective cathode electrolyte interface (CEI) film is formed on LiMn_2O_4 from the preferential oxidation of TPB, which suppresses the oxidation decomposition of the standard electrolyte. Due to the incorporation of boron, the CEI film formed from TPB is beneficial to the rate capability of LiMn_2O_4 .

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1. Introduction

Over the past decades, lithium ion battery has dominated the power source market for small portable electronic devices due to its higher energy density, longer cycle and calendar life than other secondary batteries. [1,2]. Nowadays, it is used as an indispensable power source for large-scale electric vehicles [3,4]. However, the cost of lithium ion battery is high compared to its counterparts such as lead-acid and nickel-hydride batteries, when it is used in electric vehicles. It is a challenge to reduce the cost for the fabrication of lithium ion battery.

Traditional cathodes such as layered lithium cobalt oxide (LiCoO_2) and olivine lithium iron phosphate (LiFePO_4) are mainly responsible for the high cost of currently commercialized lithium ion battery, due to the deficient resource of cobalt and the complicate synthesis of LiFePO_4 [4–11]. Comparatively, spinel lithium manganese oxide (LiMn_2O_4) attracts much attention due to

the abundant resource of manganese, the easiness for its synthesis and its environmental friendliness [12,13]. However, the interfacial instability between LiMn_2O_4 and conventional carbonate-based electrolyte, especially under elevated temperature, limits its application in practice [14–17]. The electrolyte tends to decompose on LiMn_2O_4 generating HF that causes the dissolution of manganese ions from LiMn_2O_4 [18–21].

A number of approaches have been developed to improve the interfacial stability between LiMn_2O_4 and electrolyte. Doping elements such as Co, Ni, Cr, or F in LiMn_2O_4 or coating LiMn_2O_4 with inert compounds such as Al_2O_3 and ZrO_2 are found to be effective for the improvement in structural stability of LiMn_2O_4 [22,23]. However, these approaches not only require complicate synthetic processes and add cost into the product LiMn_2O_4 , but also reduce the specific capacity of LiMn_2O_4 due to the introduction of inert components.

Based on the successful application of electrolyte additives from maintaining structural integrity of graphite anode of lithium ion battery, electrolyte additives have also been used to form cathode electrolyte interphase (CEI) films to improve the interfacial stability of various cathodes in carbonate-based electrolyte [24–26]. Similarly to the coating with inert compounds, the CEI

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films formed from electrolyte additives separate cathodes from direct contact with electrolyte, and thus suppress the oxidation decomposition of electrolyte and the dissolution of transition metal ions from cathode. Comparatively, the approach by applying an electrolyte additive for the interfacial stability improvement of cathode is simple and cost-efficient.

Several electrolyte additives including prop-1-ene-1,3-sultone (PES) [27], 3,3'-sulfonyldipropionitrile [28] and dimethyl phenylphosphonite (DMPP) [29], have been developed for improving the interfacial stability of LiMn_2O_4 in carbonate-based electrolyte. Unfortunately, these additives yield adverse effect on the electrochemical performance of LiMn_2O_4 when they are used in a higher concentration. For example, the discharge capacity of LiMn_2O_4 is decreased from the 113 mAh g^{-1} in the base electrolyte to 106 mAh g^{-1} and 102 mAh g^{-1} for 0.5% and 1% DMPP-containing electrolytes, respectively [29]. This adverse effect is attributed to the poor ionic conductivity of the CEI film formed from the additives, which is detrimental to the rate capability of LiMn_2O_4 .

Boron-containing compounds are often used as electrolyte additives for the formation of protective films for various cathodes of lithium ion battery [25,30–35]. Differently from other electrolyte additives, boron-containing compounds form CEI films with lower interfacial impedance that is independent of the concentration of the additives. This feature of boron-containing compounds is attributed to the coordination of boron with F^- ions, which reduces the formation of highly resistive LiF [36–39]. However, most of boron-containing additives for the formation of CEI film have complicated molecule structures, such as tris(trimethylsilyl) borate (TMSB) [25], tris(pentafluorophenyl) borane (TPFPB) [37] and tris(2H-hexafluoroisopropyl) borate (THFPB) [40], which may decrease the ionic conductivity of the electrolyte. Boron-containing salts such as lithium bis(oxalate)borate (LiBOB) were also used for the cyclic stability improvement of lithium ion batteries, but their contributions result mainly from the thermal stability improvement of the electrolyte due to their substitutions for the less stable of LiPF_6 [41].

It has been found that some simple boron-containing compounds such as trimethyl borate (TMB) and triethyl borate (TEB) are effective as good as TMSB for improving the interfacial stability of different cathodes but hardly affected the properties of the electrolyte even if they are used in high concentrations [30,42,43]. In this work, we reported a new finding that another simple boron-containing compound, tripropyl borate (TPB), was effective for stabilizing the interface between LiMn_2O_4 and carbonate-based electrolyte under elevated temperature. At 55°C , not only the cyclic stability of LiMn_2O_4 was improved significantly, but also the self-discharge of the charged LiMn_2O_4 was suppressed effectively. The contribution of TPB was understood through electrochemical measurements, theoretical calculations and physical characterizations. The electrochemical measurements and physical characterizations demonstrated that TPB functions like other simple boron-containing electrolyte additives such as the same category TMB and TEB, forming a protective CEI on cathode. However, theoretical calculations indicated that TPB had a lower oxidation potential than TMB and TEB, which is beneficial for the formation of an effectively protective CEI film [31].

2. Experimental

2.1. Preparation

Tripropyl borate (TPB) (99%) was purchased from Aladdin Industrial Inc, China, and used without further purification. The standard (STD) electrolyte (1.0M LiPF_6 in a mixed solvent of ethylene carbonate (EC)/ethyl methyl carbonate (EMC)/diethyl

carbonate (DEC) (3/5/2, in weight)) was provided by Dongguan Kaixin Materials Technology Co. Ltd, China. Various contents of TPB were added into the STD electrolyte in weight percentage to obtain 2.0, 3.0 and 4.0 wt. % TPB-containing electrolytes. All the electrolytes were prepared in a highly pure argon-filled glove box (MBRAUN, Germany), in which water and oxygen contents were controlled to less than 0.1 ppm. The LiMn_2O_4 electrodes were prepared by blending 80 wt. % LiMn_2O_4 (Qingdao Xinzhen Lithium industry Co., Ltd, China), 10 wt. % Acetylene Black and 10 wt. % polyvinylidene fluoride (PVDF) in N-methylpyrrolidone (NMP), coating the slurry on aluminum foil, drying under vacuum, and punching into disks of 1.44 cm^2 . To ensure the consistency of all the electrodes, the loading of LiMn_2O_4 in the electrodes were controlled at the level of $1.81 \pm 0.01 \text{ mg cm}^{-2}$. Based on the electrode area (1.44 cm^2), the loading of LiMn_2O_4 in the electrodes, and the theoretical specific capacity of LiMn_2O_4 (148 mAh g^{-1}), the theoretical discharge capacity of the prepared LiMn_2O_4 electrodes was $0.39 \pm 0.01 \text{ mAh}$. To evaluate the effect of the additive on anode of lithium ion battery, graphite electrodes were prepared by using the mixture of 80 wt. % artificial graphite (provided by Dongguan Kaixin New Energy Technology Co., Ltd, China), 10 wt. % acetylene black and 10 wt. % polyvinylidene fluoride (PVDF). Li/ LiMn_2O_4 or Li/graphite 2025-coin cell for electrochemical measurement was assembled using Celgard 2400 as separator and lithium foil as counter and reference electrode in the glove box mentioned above.

2.2. Electrochemical measurements

The cycle stability and rate capability of LiMn_2O_4 electrode were evaluated via constant current charge/discharge tests in Li/ LiMn_2O_4 coin cell between 3.0 to 4.5 V on LAND test system (CT2001A, China). In the assessment of cycle stability, the cell was charged and discharged at 0.5C ($1\text{C} = 148 \text{ mAh g}^{-1}$) for the first three cycles and then at 1C for the subsequent cycles. In the assessment of rate capability, the cell also was charged and discharged at 0.5C for the first three cycles and then at various currents of 1C, 4C, 6C, and 10C. The Li/graphite cells were charged and discharged with 0.1C ($1\text{C} = 372 \text{ mAh g}^{-1}$) for the first two cycles and 0.3C for the subsequent cycles at room temperature. Self-discharge test was performed after the charging the Li/ LiMn_2O_4 cell to 4.5 V, and then open-circuit voltage (OCV) was traced at 55°C . The electrochemical impedance spectroscopy was performed on an Autolab (PGSTAT302N) electrochemical station in a frequency range from 10^5 to 0.1 Hz with the amplitude of 5 mV. To ensure the accuracy of the data, all electrochemical measurements on each sample were performed with five cells. The reported data are those from the cell whose values are the mean ones among five cells.

2.3. Physical characterizations

For characterizing the cycled LiMn_2O_4 electrodes, the cells were disassembled in the glove box. The cycled LiMn_2O_4 electrodes were rinsed with anhydrous DMC three times to remove residual solvents and LiPF_6 salt on the surface and then dried over night at room temperature. The surface morphology of the cycled electrodes was observed by scanning electron microscopy (SEM, JSM-2100, Japan) and transmission electron microscopy (TEM, JEM-2100HR, JOEL, Japan). The surface compositions of the LiMn_2O_4 electrodes were analyzed by X-ray photoelectron spectroscopy (XPS, ESCALAB 250, US) by using a focused monochromatized Al $K\alpha$ radiation ($h\nu = 1486.6 \text{ eV}$) as the X-ray source under ultra-high vacuum.

2.4. Calculation methods

All calculations were carried out using the Gaussian 09 package [44]. The geometric structures were optimized with the B3LYP in conjunction with the 6–311++G (d) basis set [45]. In addition, frequency calculations were performed at the same basis set to confirm that each optimized structure. In order to investigate the effects of bulk solvents, the structures are optimized by using the polarized continuum model (PCM), with acetone (a dielectric constant of 20.5) as the default solvent for all PCM calculations. The calculated oxidation potential (E_{ox}) was converted from the absolute oxidation potential of the solvents (vs. Li/Li⁺) [46,47]:

$$E_{ox}(\text{Li}^+/\text{Li}) = [G(\text{M}^+) - G(\text{M})]/F - 1.4 \text{ V} \quad (1)$$

where $G(\text{M})$ and $G(\text{M}^+)$ is the free energy of the electrolyte component M and its oxidized form M⁺ at 298.15 K, respectively, and F is the Faraday constant.

3. Results and discussion

3.1. Cyclic stability improvement and self-discharge suppression

To evaluate the influence of TPB on the cyclic stability of LiMn₂O₄, constant current charge/discharge test was performed on Li/LiMn₂O₄ coin cells. The obtained results are presented in Fig. 1. As shown in Fig. 1a, LiMn₂O₄ electrode in STD electrolyte suffers a

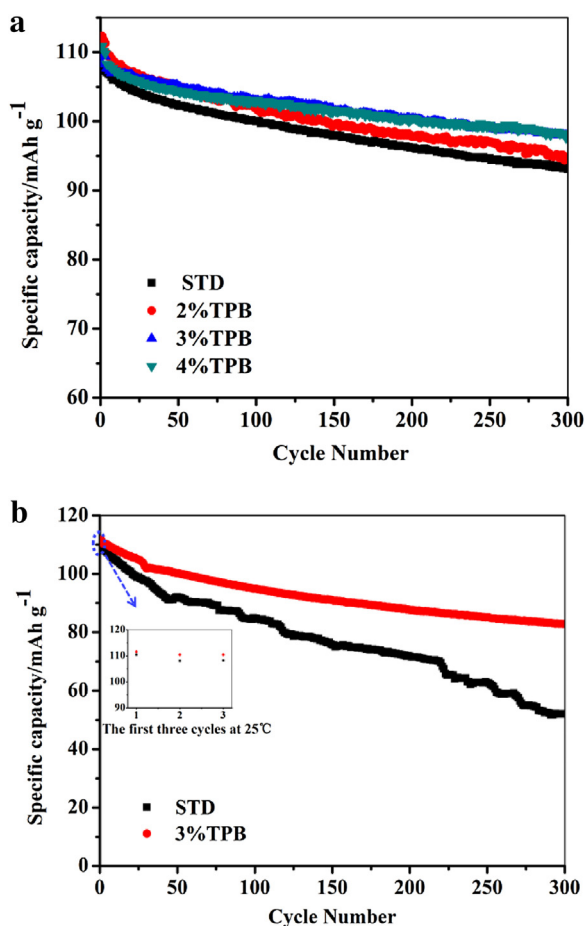


Fig. 1. Cyclic stability of LiMn₂O₄ electrode in STD and TPB-containing electrolytes under 25 °C at 0.5C for the first three cycles and at 1C for the subsequent cycles (a), and under 25 °C at 0.5C for the first three cycles and then under 55 °C at 1C for the subsequent cycles (b) in the potential range of 3.0–4.5 V.

capacity loss when it is performed with cycling at room temperature. The capacity retention is 85% after 300 cycles, indicative of the instability of LiMn₂O₄/electrolyte interface, which mainly results from the electrolyte decomposition and manganese dissolution [48]. When TPB is applied, the cyclic stability of LiMn₂O₄ is improved, showing that TPB can suppress electrolyte decomposition and manganese dissolution. With increasing the content of TPB from 0% to 3%, the capacity retention of LiMn₂O₄ rises to 90% after 300 cycles, indicating that 3% TPB is effective for the cyclic stability improvement of LiMn₂O₄. It should be noted that the cyclic performances of LiMn₂O₄, in terms of discharge capacity and cyclic stability, are hardly affected by the further increased the content of TPB. Apparently, increasing the content of TPB does not yield adverse effect on the ionic conductivity of the electrolyte, which should be attributed to its smaller molecule than other boron-containing compound such as TMSB. Since the best performance of LiMn₂O₄ can be achieved by using 3% TPB, this content was considered for further investigations.

The effectiveness of TPB can be indicated more clearly at elevated temperature. Fig. 1b compares the cyclic performance of LiMn₂O₄ electrode in STD and 3% TPB-containing electrolytes under 55 °C at 1C rate in the potential range of 3.0–4.5 V. Before the cycling test under 55 °C, the electrode was performed at 0.5C for three cycles under room temperature in the potential range of 3.0–4.5 V. Differently from the cycling behavior under room temperature (Fig. 1a), the LiMn₂O₄ cycled in STD electrolyte shows a fast capacity decay under 55 °C, with a capacity retention of only 47% after 200 cycles. This difference suggests that the high temperature is detrimental to the interfacial stability of LiMn₂O₄/electrolyte, where electrolyte decomposition and manganese dissolution will take place more dramatically. When 3% TPB is added into the STD electrolyte, the capacity retention of LiMn₂O₄ is improved to 74%. This value is comparable to that achieved by using DMPP, the best electrolyte additive that has been reported in literature [29]. With 0.5% DMPP added into a base electrolyte (1.0 M LiPF₆ in EC/EMC (3/7, in volume), the capacity retention of LiMn₂O₄ was improved to 82% after 200 cycles at 1C under 55 °C. Apparently, TPB is effective for the cyclic stability improvement of LiMn₂O₄ under elevated temperature.

The interfacial instability of LiMn₂O₄/electrolyte during cycling is mainly caused by electrochemical oxidation decomposition of the electrolyte. There might also exist a chemical process between the charged LiMn₂O₄ and electrolyte during storage whereby the

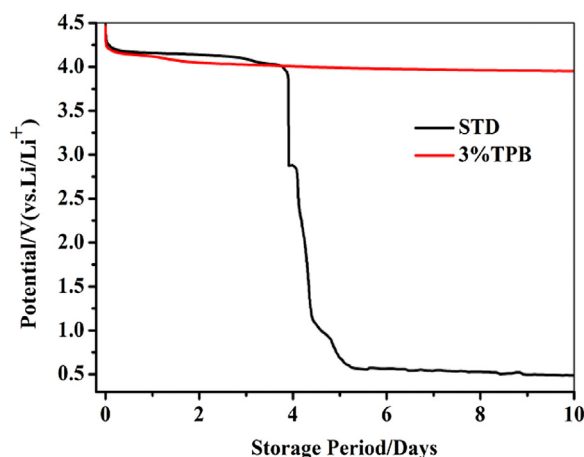


Fig. 2. Potential variation of the charged LiMn₂O₄ electrode in STD and 3% TPB-containing electrolytes after three cycles at 0.5C under room temperature and then storage under 55 °C.

charged capacity is lost, which is called as self-discharge. To evaluate the influence of TPB on the self-discharge of charged LiMn_2O_4 electrode in electrolyte, the potential variation of charged LiMn_2O_4 electrode was recorded when it was stored under 55°C . Before storage, the LiMn_2O_4 electrode was performed with three charge/discharge cycles at 0.5C under room temperature in the potential range of 3.0–4.5 V at 25°C . As shown in Fig. 2, the potential of the charged LiMn_2O_4 electrode in the STD electrolyte remains relatively stable initially but drops quickly after 4 hours' storage (from 4.0 V to 0.5 V), suggesting that the charged LiMn_2O_4 suffers a serious self-discharge in the STD electrolyte. By contrast, the potential of the charged LiMn_2O_4 electrode in 3% TPB-containing electrolyte maintains its relatively stable potential up to the end of storage test, suggesting that the self-discharge that happens in the STD electrolyte has been suppressed successfully.

In the STD electrolyte, the carbonate electrolytes tend to be oxidized by the charged LiMn_2O_4 , especially at elevated temperature. Accordingly, the Mn^{4+} on the surface of the charged LiMn_2O_4 are reduced to Mn^{3+} , corresponding to the potential plateaus at 4.0–4.2 V, which is accompanied with lithium ion insertion into the tetrahedral sites of the spinel, similarly to the initial discharge process during cycling, which does not involve any phase transition [49]. When the reaction of the charged LiMn_2O_4 with electrolyte proceeds further, however, the inserted lithium ions might take the octahedral sites of the spinel. This process corresponds to the potential plateau at about 2.7 V, which involves a cubic to tetragonal phase transition [50,51]. Besides, the formed Mn^{3+} might cause crystallographic distortion due to Jahn-Teller effect and manganese dissolution through its disproportionation at

high temperature [52]. Therefore, the LiMn_2O_4 on the surface suffers a structural destruction. Furthermore, the products from the successive electrolyte decomposition on LiMn_2O_4 together with new compounds formed from the lithium insertion isolate the bulk spinel from current collector or carbon conductor. These effects may cause the exposure of the current collector (aluminum) to the electrolyte and the dissolution of aluminum oxide on the current collector, which corresponds to the potential plateau at about 1.0 V. The dissolution of aluminum oxide causes the exposure of aluminum, leading to the final potential as low as 0.5 V. The stable potential of the charged LiMn_2O_4 in 3% TPB-containing electrolyte indicates that the self-discharge reactions that take place in the STD electrolyte have been successfully suppressed.

The cyclic stability improvement and the self-discharge suppression of LiMn_2O_4 in TPB-containing electrolyte obviously indicate that TPB is able to stabilize the interface of LiMn_2O_4 /carbonate electrolyte. This stabilization should be attributed to the protective CEI film on LiMn_2O_4 formed from the preferential oxidation of TPB, which suppresses the electrolyte decomposition and protects LiMn_2O_4 from structural destruction.

3.2. Preferential oxidation of TPB

The preferential oxidation of TPB on LiMn_2O_4 was confirmed by theoretical calculations and charge behavior analyses. Since the oxidation of solvents or additives usually takes place in the form of the coordinates with the anion PF_6^- of salt in the electrolyte [53], the oxidation potentials of solvents and TPB in single and

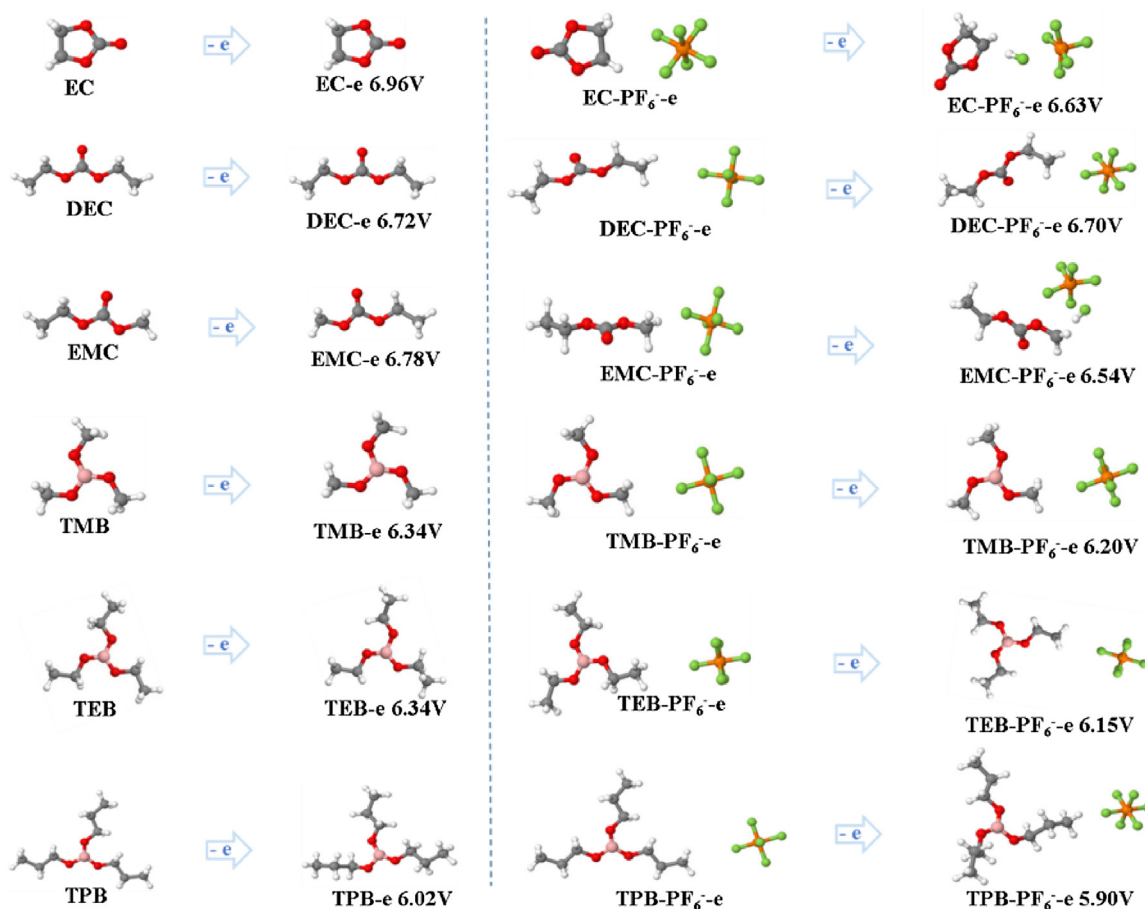


Fig. 3. Optimized structures and calculated oxidation potential (in V vs. Li/Li^+) of EC, DEC, EMC, TMB, TEB, and TPB, and their anion clusters.

coordinated form were calculated, with a comparison of two simple boron-containing electrolyte additives, TMB and TEB. The obtained results are presented in Fig. 3. It can be seen from Fig. 3 that TPB has a lower oxidation potential than the solvents, EC, DEC or EMC, whether in single or in the form of coordinates with anion. These calculations suggest that TPB would oxidize preferentially compared to the STD electrolyte. Additionally, TPB has a lower oxidation potential than TMB and TEB, which is the reason why TPB is selected in this study. Since the CEI film is formed from the oxidation of the additives or the electrolyte, the preferential oxidation of the additives might mitigate the interference from electrolyte decomposition during the formation of the CEI film.

In fact, the preferential oxidation of TPB can be indicated by comparing the initial charge profiles of LiMn_2O_4 electrodes in STD and TPB-containing electrolytes, as shown in Fig. 4. Fig. 4a presents the initial charge/discharge profiles at 0.5C under room temperature. It can be seen from Fig. 4a that there appears a potential change at 3.8–4.0 V for the electrode in 3% TPB-containing electrolyte compared to the smoothly rising potential for the electrode in the STD electrolyte, indicative of the preferential oxidation of TPB to the STD electrolyte. This preferential oxidation can be more clearly observed in dQ/dV curves. As shown in Fig. 4b, both electrodes are characteristic of LiMn_2O_4 with a couple of redox peaks corresponding to the typical lithium insertion/extraction reaction in spinel. Differently from the electrode in the STD electrolyte, the electrode in 3% TPB-containing electrolyte shows an additional significant oxidation peak at about 3.93 V. This

peak potential is lower than that for the lithium extraction and those for the electrolyte decomposition. Therefore, TPB is oxidized at the potential before lithium extraction and electrolyte decomposition. The preferential oxidation of TPB is important for the formation of a protective CEI film on LiMn_2O_4 , because the interference from the volume change due to the lithium extraction and the electrolyte decomposition products can be avoided [31,51].

3.3. CEI film formed by TPB

The existence of the CEI film formed by TPB can be confirmed by physical characterizations including SEM, TEM and XPS. Fig. 5 presents the SEM and TEM images of the fresh LiMn_2O_4 electrode and the cycled ones in STD and TPB-containing electrolytes. It can be seen from Fig. 5a and d that the spinel morphology and the clean boundary can be observed on the fresh LiMn_2O_4 particles. Whereas, the cycled sample in STD electrolyte, as shown in Fig. 5b and e, presents absolutely different morphology from the fresh one: the spinel morphology of LiMn_2O_4 particles almost disappears (Fig. 5b) and the dark TEM image of the fresh LiMn_2O_4 particle turns to be bright (Fig. 5e). Additionally, there appear thick and uneven deposits on the cycled LiMn_2O_4 . The morphological change of LiMn_2O_4 particles should be ascribed to the structural collapse and the thick and uneven deposits are attributed to the electrolyte decomposition products, which account for the poor cycling stability of LiMn_2O_4 when is cycled in the STD electrolyte under elevated temperature. By contrast, the cycled LiMn_2O_4 in the TPB-containing electrolyte maintains the morphological integrity and a uniform and thin CEI film can be clearly identified, as shown in Fig. 5c and e. These observations confirm that a protective CEI film has been successfully formed on LiMn_2O_4 , which stabilizes the LiMn_2O_4 /electrolyte interface through suppressing electrolyte decomposition.

The surface chemical composition of the cycled electrodes in the STD and TPB-containing electrolyte were analyzed by XPS. The O1s, F1s, B1s, and Mn2p spectra of both electrodes are shown in Fig. 6. In the O1s spectra, the peaks, located at 529.9, 531.8 and 533.5 eV, are corresponding to metal oxide (M–O) in Mn_xO_y or LiMn_2O_4 , Li_2CO_3 and C–O in lithium alkyl carbonates ($\text{R-CH}_2\text{OCO}_2^-$ Li) or polycarbonates, respectively [54,55]. In the F1s spectra, the electrode cycled in STD electrolyte show the far stronger LiF peak than that in TPB-containing electrolyte. In the Mn2p spectra, three main peaks can be identified, corresponding to Mn^{3+} (642.1 eV), Mn^{4+} (644.0 eV) and $\text{Mn}2p^{1/2}$ (654.1 eV), respectively [56,57], and the corresponding peak intensity on electrode cycled in STD electrolyte is weaker than that in TPB-containing electrolyte. The weaker M–O and Mn peaks and the stronger Li_2CO_3 and LiF peaks on the electrode cycled in STD electrolyte confirm that the deposits from electrolyte decomposition in the STD electrolyte is thicker than the CEI film formed by TPB. LiF existing in the deposit will increase the interfacial impedance [42]. It should be noted that B–O peak can be identified on the electrode cycled in TPB-containing electrolyte but does not exist on the electrode cycled in STD electrolyte. Obviously, the CEI film on LiMn_2O_4 cycled in TPB-containing electrolyte consists mainly of the oxidation decomposition products of TPB and thus provides LiMn_2O_4 with an effective protection.

The boron-containing CEI film stabilizes the LiMn_2O_4 /electrolyte interface, which is embodied by the less polarization, the smaller interfacial impedance and the suppression of electrolyte decomposition. Fig. 7 compares the electrochemical performances of LiMn_2O_4 electrode in STD and TPB-containing electrolytes, in terms of discharge voltage plateaus, interfacial impedance and rate capability. It can be found from Fig. 7a and b that the medium discharge voltage of $\text{Li/LiMn}_2\text{O}_4$ cell in STD electrolyte decreases more seriously with cycling, dropping from 4.1 to 3.7 V after 300

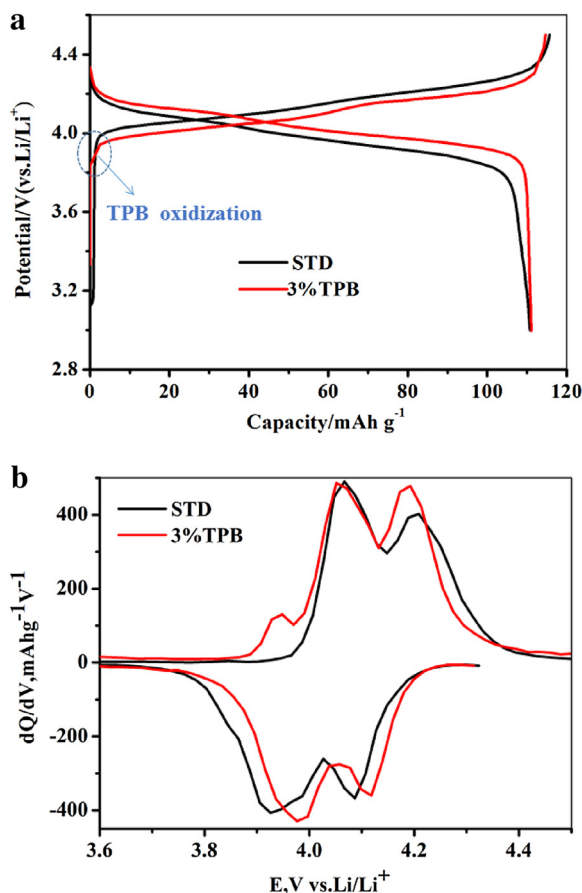


Fig. 4. The first charge and discharge curves (a) and corresponding differential capacity curves (b) of LiMn_2O_4 electrode in STD and 3% TPB-containing electrolytes under 25 °C. The charge and discharge rate current was 0.5C in the potential range of 3.0–4.5 V.

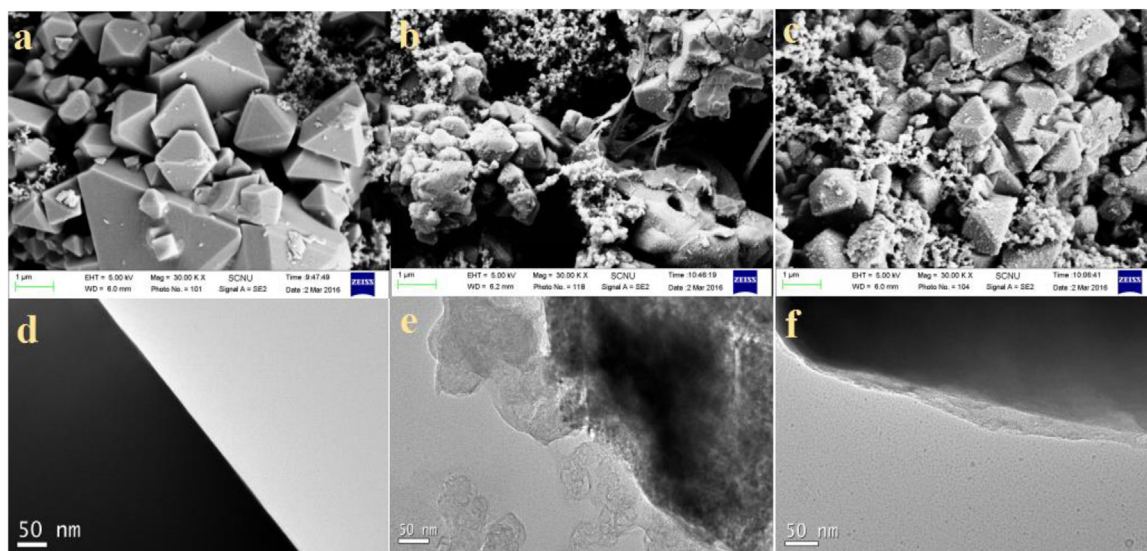


Fig. 5. SEM and TEM images of fresh (a, d) and cycled LiMn₂O₄ in STD (b, e) and 3% TPB-containing (c, f) electrolytes.

cycles with a change of 0.4 V compared to the 0.06 V in TPB-containing electrolyte, indicative of the reduced polarization due to the CEI film formed from TPB.

The electrochemical impedance spectra presented in Fig. 7c and d consist of the overlapped semicircles in the high-to-medium frequency and a slope line at low frequencies. The spectra were obtained from Li/LiMn₂O₄ cells. Since the redox reaction of Li/Li⁺ on the lithium anode is far fast compared to the lithium insertion/extraction on LiMn₂O₄ cathode, the contribution of lithium anode can be neglected and the spectra of the cells represent the interfacial impedance of LiMn₂O₄ cathode [25]. The overlapped semicircles include the impedance for the ion transportation in CEI film and the charge transfer on the interface, while the slope line at low frequencies corresponds to the Warburg impedance [58]. Initially (after three cycles), the electrode in TPB-containing electrolyte has smaller interfacial resistance (the diameters of

the semicircles) than that in STD electrolyte: about 50 Ω in the TPB-containing electrolyte compared to the over 200 Ω in the STD electrolyte, indicative of the ionically conductive ability of the CEI film formed from TPB. After 300 cycles, the impedance increases significantly for the electrode in STD electrolyte (from 200 Ω to 800 Ω) but insignificantly for that in TPB-containing electrolyte (from 50 Ω to 70 Ω), suggesting that the interface of LiMn₂O₄/electrolyte collapses in STD electrolyte but can be stabilized by applying TPB. The successive oxidation decomposition of electrolyte on LiMn₂O₄ in the STD electrolyte yields the accumulation of insulative Li₂CO₃ and LiF that contributes the increased interfacial impedance and erosive HF that causes the manganese dissolution, leading to the collapse of LiMn₂O₄/electrolyte interface. With applying TPB, a protective CEI film is formed and the LiMn₂O₄/electrolyte interface is stabilized.

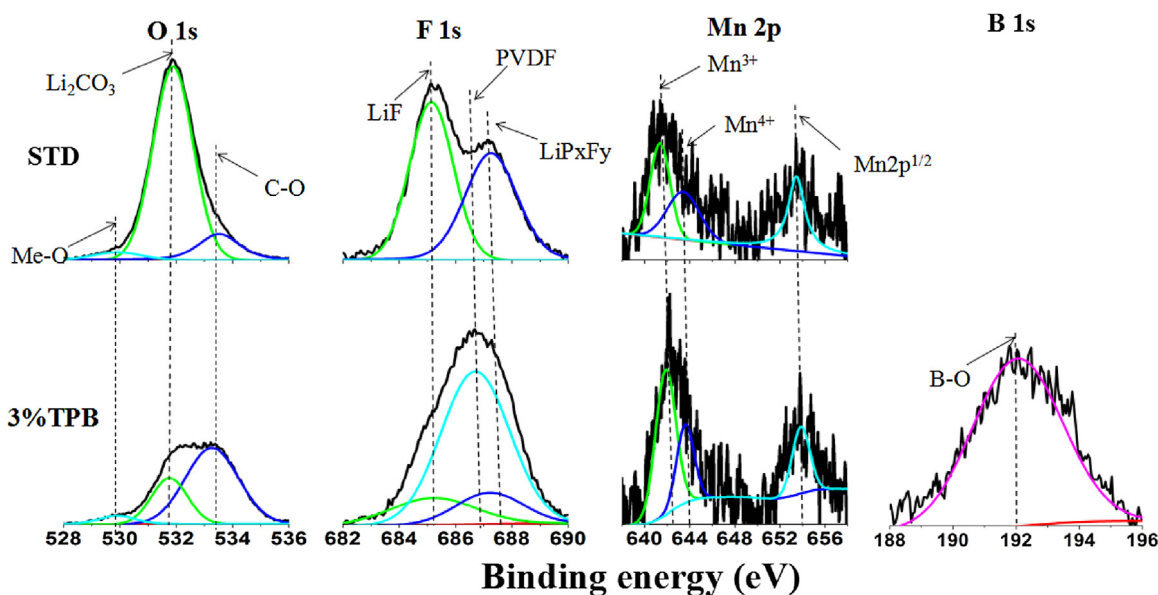


Fig. 6. XPS profiles of cycled LiMn₂O₄ in standard and 3% TPB-containing electrolytes.

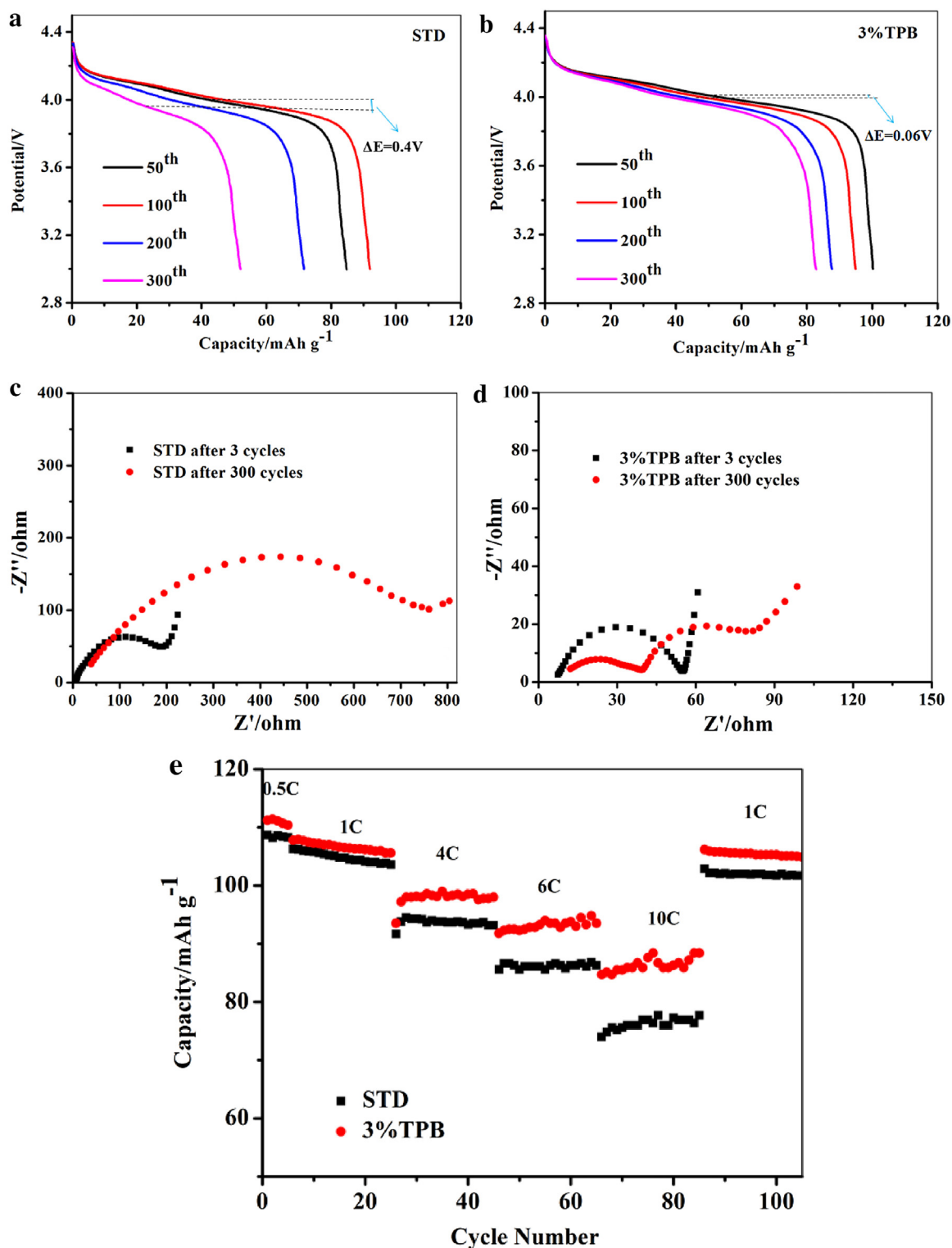


Fig. 7. Selected discharge profiles (a, b), electrochemical impedance spectra (c, d) in a frequency range from 10^5 to 0.1 Hz with the amplitude of 5 mV and rate capability (e) of LiMn_2O_4 electrode in STD and 3% TPB-containing electrolytes.

The lower interfacial impedance of LiMn_2O_4 in TPB-containing electrolyte would be beneficial to its rate capability. Fig. 7e presents discharge capacity of LiMn_2O_4 electrodes in STD and TPB-containing electrolytes under various rate currents. It can be found from Fig. 7e that the LiMn_2O_4 electrode in 3% TPB-containing electrolyte always delivers larger discharge capacity than that in STD electrolyte, especially at higher rate current. At 10C, the

average discharge capacity of LiMn_2O_4 in TPB-containing electrolyte is 87 mAh g^{-1} , compared to the 10 mAh g^{-1} in STD electrolyte. Therefore, the CEI film formed by TPB benefits the capacity delivery of LiMn_2O_4 , especially under high rate current.

The suppression of electrolyte decomposition by the CEI film formed from TPB can be observed by comparing the responding current under a high potential. Fig. 8 shows the

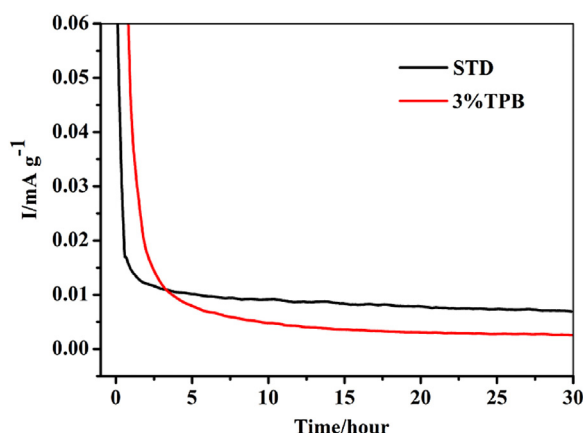


Fig. 8. Chronoamperometric responses of LiMn_2O_4 electrode maintained at 4.5 V for 30 hours under 55 °C after three cycles under 25 °C at 0.5C in the STD and 3% TPB-containing electrolytes.

chronoamperometric responses of LiMn_2O_4 electrodes that are maintained at 4.5 V under 55 °C in STD and TPB-containing electrolytes. Before the chronoamperometry, the electrodes were performed at 0.5C under 25 °C for three charge/discharge cycles. It can be found from Fig. 8 that a large residue current can be recorded on the electrode in STD electrolyte but the residue current decreases gradually to zero in the TPB-containing electrolyte. At 4.5 V, the carbonate-based electrolyte is unstable and decomposes seriously, especially under elevated temperature, leading to the large residue current on the electrode in STD electrolyte. In the TPB-containing electrolyte, however, a CEI film is formed due to the preferential oxidation of TPB, which separates LiMn_2O_4 from direct contact with electrolyte, and therefore the electrochemical oxidation decomposition reactions of the electrolyte on LiMn_2O_4 and the self-discharge reactions between charged LiMn_2O_4 and electrolyte are suppressed.

Usually, an additive for cathode performance improvement might yield a negative effect on the anode of lithium ion battery. To understand the effect of TPB on graphite anode, Li/graphite cells were assembled using STD and 3% TPB-containing electrolytes and performed with charge/discharge test. Fig. 9 presents the cycling performances of graphite anode in STD and 3% TPB-containing electrolytes. It can be found that both electrodes have almost the

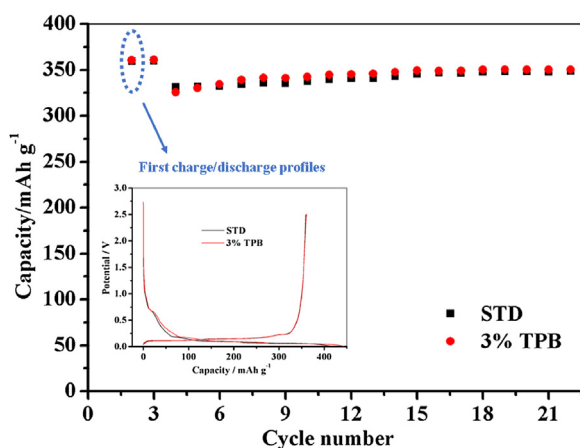


Fig. 9. Cyclic performance of graphite electrode under 25 °C at 0.1C for the first two cycles and 0.3C for the subsequent cycles in STD and 3% TPB-containing electrolytes.

same initial charge/discharge profiles and the same cyclic performance. This comparison indicates that TPB does not yield any negative effect on graphite anode.

4. Conclusions

Trimethyl borate (TPB) is able to stabilize the LiMn_2O_4 /carbonate-based electrolyte interface. Theoretical calculations and electrochemical analyses indicate that TPB oxidizes preferentially to the electrolyte. Further electrochemical measurements and physical characterizations demonstrate that a CEI film is formed due to the preferential oxidation of TPB. The resulting CEI film suppresses the electrolyte decomposition that yields unnecessary products such as LiF and HF, and therefore the LiMn_2O_4 /carbonate-based electrolyte interface is stabilized. With this stabilized interface, the cyclic stability and rate capability of LiMn_2O_4 is improved and the self-discharge of the charged LiMn_2O_4 is suppressed significantly, especially under elevated temperature. Since TPB is a simple boron-containing molecule, its application does not yield any adverse effect on the properties of carbonate-based electrolyte.

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